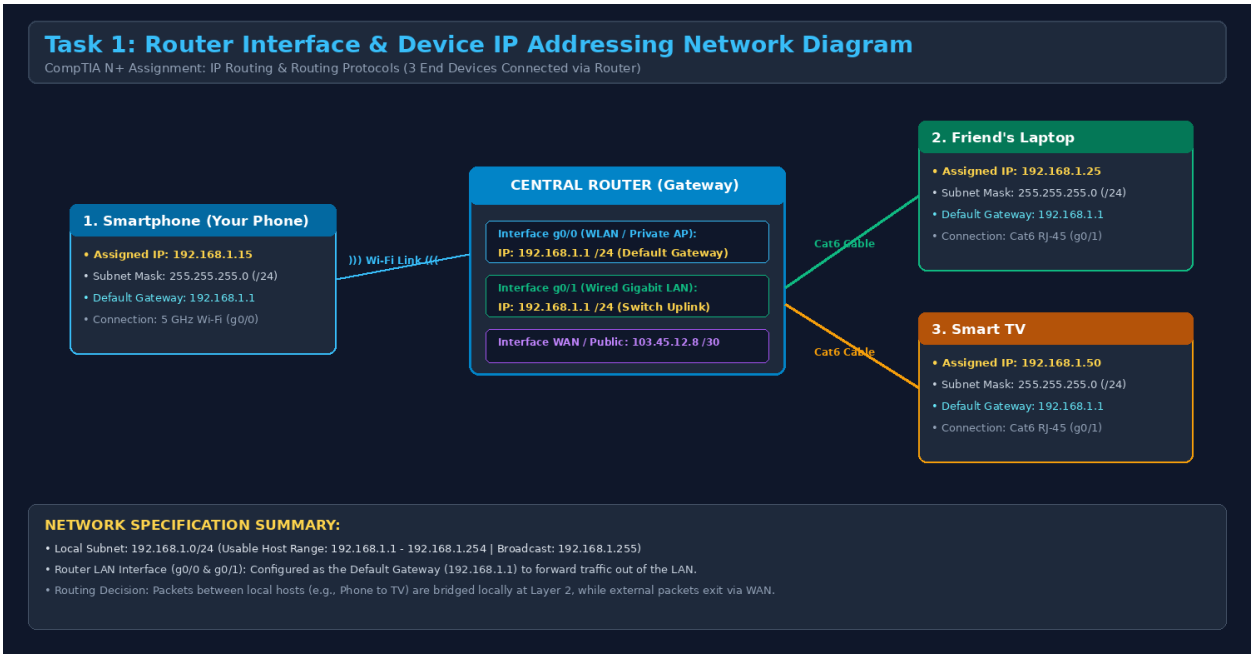
IP routing & IP protocols

Task 1: Network Diagram (Router Interfaces & Device IP Addressing)

The diagram below maps the three client devices connected through a central gateway router:



Device & Interface Configuration Table:

- **Router LAN Gateway Interfaces (g0/0 & g0/1):** IP 192.168.1.1 (Subnet Mask: 255.255.255.0 / /24)
- **Router WAN Interface:** IP 103.45.12.8 (Public Gateway / /30)
- **Device 1 (Your Smartphone):** IP 192.168.1.15 (Connected via 5 GHz Wi-Fi)
- **Device 2 (Friend's Laptop):** IP 192.168.1.25 (Connected via Cat6 Ethernet cable)
- **Device 3 (Smart TV):** IP 192.168.1.50 (Connected via Cat6 Ethernet cable)
- *Common Parameters:* All three devices share Default Gateway 192.168.1.1 and Subnet Mask 255.255.255.0.

Task 2: Step-by-Step Simulation of Manual IP Routing Across Subnets

Network Scenario & IP Setup:

- **Source (Your Laptop):** 192.168.10.15 /24 (Default Gateway: 192.168.10.1)
- **Router 1 (Local Router - R1):**
    - Interface g0/0 (LAN): 192.168.10.1 /24
    - Interface s0/0/0 (WAN Point-to-Point Link): 10.0.0.1 /30
- **Router 2 (Remote Router - R2):**
    - Interface s0/0/0 (WAN Point-to-Point Link): 10.0.0.2 /30
    - Interface g0/0 (Remote LAN): 192.168.20.1 /24
- **Destination (Friend's PC):** 192.168.20.50 /24 (Default Gateway: 192.168.20.1)

[Your Laptop] 192.168.10.15

|

(LAN: g0/0: 192.168.10.1)

[Router 1]

(WAN: s0/0/0: 10.0.0.1)

| (WAN Link: 10.0.0.0/30)

(WAN: s0/0/0: 10.0.0.2)

[Router 2]

(LAN: g0/0: 192.168.20.1)

|

[Friend's PC] 192.168.20.50

Step-by-Step Data Flow & Routing Decisions:

1. **Step 1: Local Subnet Check on Source Laptop**
    - Your laptop evaluates the destination address (192.168.20.50) against its own subnet mask (255.255.255.0).
    - Because 192.168.20.0 ≠\ne⍰= 192.168.10.0, the laptop determines the destination is on a **remote network**.
    - It wraps the IP packet inside an Ethernet frame with the destination MAC address of its **Default Gateway (192.168.10.1 - R1)** and sends it out.
2. **Step 2: Router 1 Routing Table Lookup**

- R1 receives the frame on interface g0/0, strips the Layer 2 header, and reads the destination IP: 192.168.20.50.
- R1 consults its routing table:

Plaintext

R1 Routing Table:

C 192.168.10.0/24 is directly connected, GigabitEthernet0/0

C 10.0.0.0/30 is directly connected, Serial0/0/0

S 192.168.20.0/24 [1/0] via 10.0.0.2, Serial0/0/0

- **Decision:** Match found for 192.168.20.0/24. R1 forwards the packet to next-hop 10.0.0.2 through exit interface Serial0/0/0.

3. **Step 3: WAN Transit to Router 2**
    - The packet traverses the serial link and arrives at R2's interface Serial0/0/0.
4. **Step 4: Router 2 Routing Table Lookup**
    - R2 reads destination IP 192.168.20.50 and checks its routing table:

Plaintext

R2 Routing Table:

C 10.0.0.0/30 is directly connected, Serial0/0/0

C 192.168.20.0/24 is directly connected, GigabitEthernet0/0

- **Decision:** The destination network 192.168.20.0/24 is **directly connected** to interface GigabitEthernet0/0.

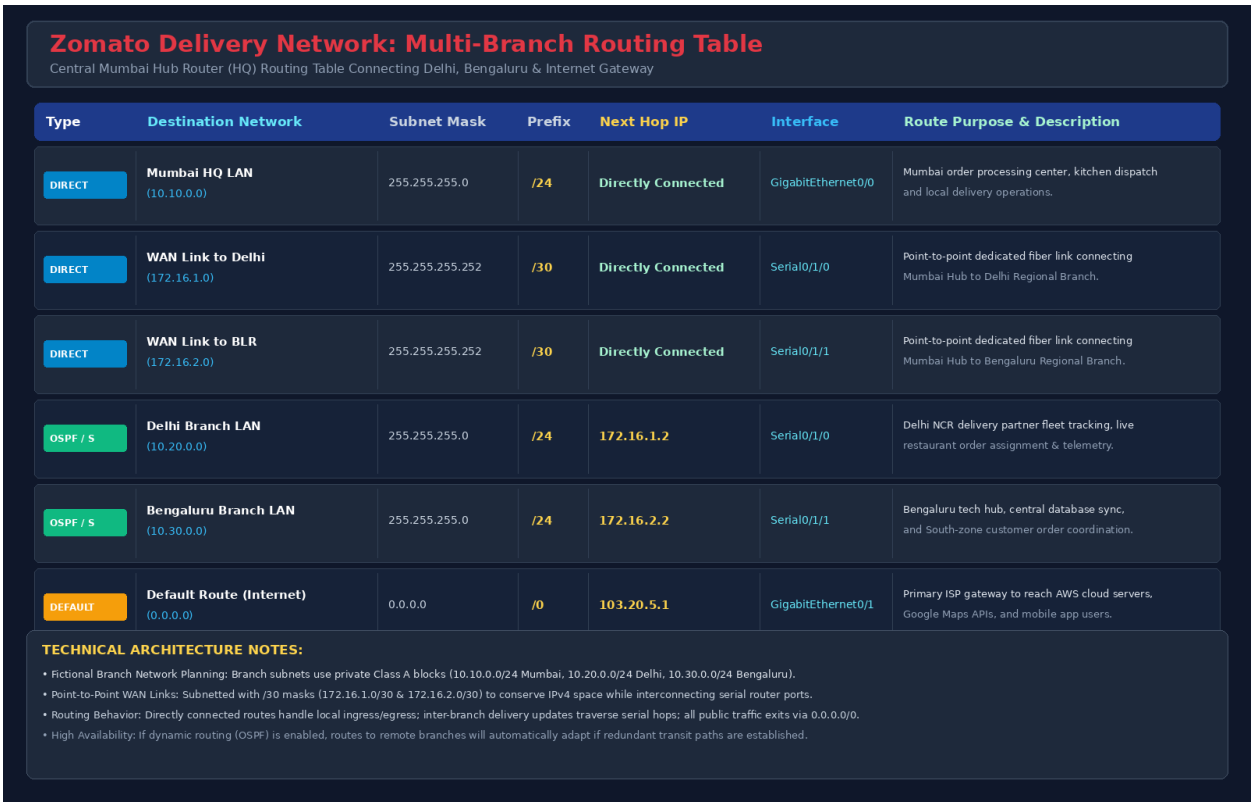
5. **Step 5: Final Delivery via ARP**
    - R2 checks its ARP table for 192.168.20.50. If not present, it broadcasts an ARP request on g0/0.
    - R2 encapsulates the IP packet into a new Ethernet frame with your friend's laptop MAC address as destination and delivers the frame.

Task 3: Routing Table for Zomato Delivery Network (3 City Branches)

Scenario Topology:

- **Headquarters Router:** Central Router in Mumbai connecting Delhi and Bengaluru regional branches.
- **Branch 1 (Mumbai Hub):** 10.10.0.0/24 (Connected to Router Interface GigabitEthernet0/0)
- **Branch 2 (Delhi Hub):** 10.20.0.0/24 (Reachable via WAN Next Hop 172.16.1.2 on Interface Serial0/1/0)
- **Branch 3 (Bengaluru Hub):** 10.30.0.0/24 (Reachable via WAN Next Hop 172.16.2.2 on Interface Serial0/1/1)
- **Internet Traffic (Cloud Orders):** Default Gateway 0.0.0.0/0 via ISP Gateway 103.20.5.1 on GigabitEthernet0/1

Routing Table (Mumbai Central Router) :



Zomato Delivery Network: Multi-Branch Routing Table

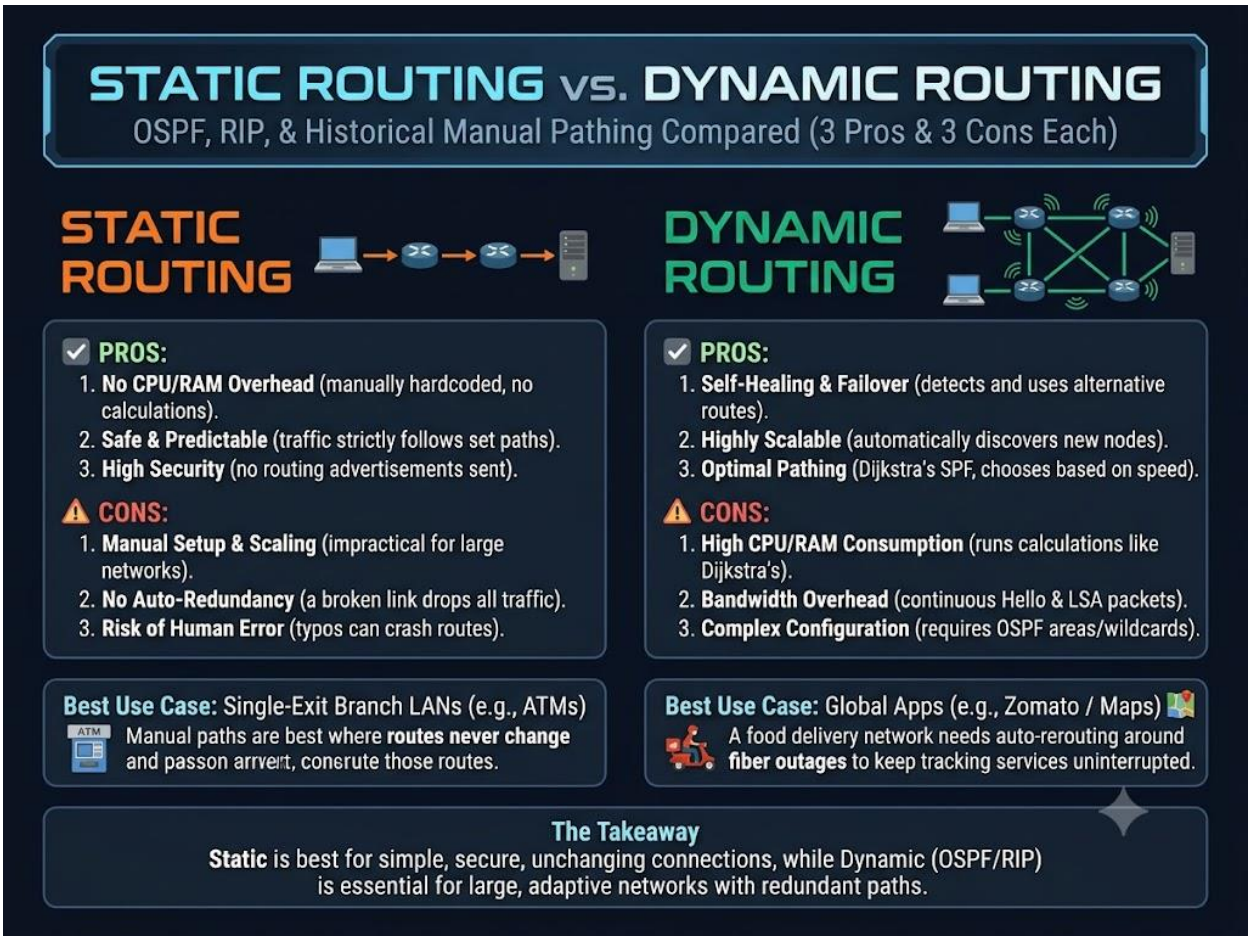
Central Mumbai Hub Router (HQ) Routing Table Connecting Delhi, Bengaluru & Internet Gateway

Type	Destination Network	Subnet Mask	Prefix	Next Hop IP	Interface	Route Purpose & Description
DIRECT	Mumbai HQ LAN (10.10.0.0)	255.255.255.0	/24	Directly Connected	GigabitEthernet0/0	Mumbai order processing center, kitchen dispatch and local delivery operations.
DIRECT	WAN Link to Delhi (172.16.1.0)	255.255.255.252	/30	Directly Connected	Serial0/1/0	Point-to-point dedicated fiber link connecting Mumbai Hub to Delhi Regional Branch.
DIRECT	WAN Link to BLR (172.16.2.0)	255.255.255.252	/30	Directly Connected	Serial0/1/1	Point-to-point dedicated fiber link connecting Mumbai Hub to Bengaluru Regional Branch.
OSPF / S	Delhi Branch LAN (10.20.0.0)	255.255.255.0	/24	172.16.1.2	Serial0/1/0	Delhi NCR delivery partner fleet tracking, live restaurant order assignment & telemetry.
OSPF / S	Bengaluru Branch LAN (10.30.0.0)	255.255.255.0	/24	172.16.2.2	Serial0/1/1	Bengaluru tech hub, central database sync, and South-zone customer order coordination.
DEFAULT	Default Route (Internet) (0.0.0.0)	0.0.0.0	/0	103.20.5.1	GigabitEthernet0/1	Primary ISP gateway to reach AWS cloud servers, Google Maps APIs, and mobile app users.

TECHNICAL ARCHITECTURE NOTES:

- Fictional Branch Network Planning: Branch subnets use private Class A blocks (10.10.0.0/24 Mumbai, 10.20.0.0/24 Delhi, 10.30.0.0/24 Bengaluru).
- Point-to-Point WAN Links: Subnetted with /30 masks (172.16.1.0/30 & 172.16.2.0/30) to conserve IPv4 space while interconnecting serial router ports.
- Routing Behavior: Directly connected routes handle local ingress/egress; inter-branch delivery updates traverse serial hops; all public traffic exits via 0.0.0.0/0.
- High Availability: If dynamic routing (OSPF) is enabled, routes to remote branches will automatically adapt if redundant transit paths are established.

Task 4: Comparison: Static Routing vs. Dynamic Routing Protocols (OSPF / RIP)



Task 5: Sample OSPF Configuration Between Two Routers & Line-by-Line Explanation

Network Setup:

- **Point-to-Point Link:** 10.0.0.0/30 between R1 (10.0.0.1) and R2 (10.0.0.2)
- **R1 LAN:** 192.168.10.0/24 | **R2 LAN:** 192.168.20.0/24
- **OSPF Process ID:** 1 | **OSPF Area:** 0 (Backbone Area)

1. Router 1 (R1) Configuration Script:

Cisco CLI

Router1# configure terminal

Router1(config)# router ospf 1

Router1(config-router)# router-id 1.1.1.1

Router1(config-router)# network 192.168.10.0 0.0.0.255 area 0

Router1(config-router)# network 10.0.0.0 0.0.0.3 area 0

Router1(config-router)# exit

2. Router 2 (R2) Configuration Script:

Cisco CLI

Router2# configure terminal

Router2(config)# router ospf 1

Router2(config-router)# router-id 2.2.2.2

Router2(config-router)# network 192.168.20.0 0.0.0.255 area 0

Router2(config-router)# network 10.0.0.0 0.0.0.3 area 0

Router2(config-router)# exit

Line-by-Line Explanation:

- **router ospf 1:**

Initializes the OSPF routing engine process on the router. The number 1 is the local Process ID (which identifies this specific instance of OSPF on this device; it does not need to match on other routers).

- **router-id 1.1.1.1 (and 2.2.2.2 on R2):**

Assigns a unique 32-bit identifier to each router in the OSPF autonomous system. This ID is used in OSPF neighbor negotiations, Designated Router (DR) elections, and database synchronization.

- **network 192.168.10.0 0.0.0.255 area 0:**

Identifies which local interfaces participate in OSPF.

  - 192.168.10.0: The network address of the LAN subnet.
  - 0.0.0.255: The **Wildcard Mask** (the inverse of subnet mask 255.255.255.0), telling OSPF to inspect all interfaces whose first 24 bits match 192.168.10.*.
  - area 0: Places this network inside the standard OSPF **Backbone Area** so route information can be shared across the autonomous system.

- **network 10.0.0.0 0.0.0.3 area 0:**

Enables OSPF on the point-to-point link connecting R1 and R2. The wildcard mask 0.0.0.3 corresponds to a /30 mask (255.255.255.252). As soon as both routers execute this command, they begin sending OSPF **Hello Packets** to multicast IP 224.0.0.5, discover each other as neighbors, establish an adjacency, and exchange routing tables automatically.